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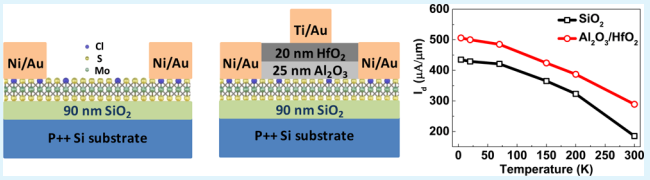
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Effect of Dielectric Interface on the Performance of MoS_2 Transistors

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ABSTRACT: Because of their wide bandgap and ultrathin body properties, two-dimensional materials are currently being pursued for next-generation electronic and optoelectronic applications. Although there have been increasing numbers of studies on improving the performance of MoS_2 field-effect transistors (FETs) using various methods, the dielectric interface, which plays a decisive role in determining the mobility, interface traps, and thermal transport of MoS_2 FETs, has not been well explored and understood. In this article, we present a comprehensive experimental study on the effect of high-k dielectrics on the performance of few-layer MoS_2 FETs from 300 to 4.3 K. Results show that Al_2O_3/HfO_2 could boost the mobility and drain current. Meanwhile, MoS_2 transistors with Al_2O_3/HfO_2 demonstrate a 2× reduction in oxide trap density compared to that of the devices with the conventional SiO_2 substrate. Also, we observe a negative differential resistance effect on the device with 1 μm-channel length when using conventional SiO_2 as the gate dielectric due to self-heating, and this is effectively eliminated by using the Al_2O_3/HfO_2 gate dielectric. This dielectric engineering provides a highly viable route to realizing high-performance transition metal dichalcogenide-based FETs.



KEYWORDS: *MoS_2, field-effect transistors, high k, pulse IV, low-frequency noise, negative differential resistance*

INTRODUCTION

After decades of ongoing progress, the silicon complementary metal−oxide−semiconductor (CMOS) industry is approaching a standstill in device performance for logic devices due to fundamental scaling limitations. To energize the future roadmap and enable continued dimensional scaling, novel materials with unique properties are being proposed to replace silicon substrate for lower supply voltage. Recently, the rich family of transition metal dichalcogenides (TMDs) MX_2, where M stands for a transition metal (Mo, W, Nb, and Ta) and X stands for a chalcogenide atom (S, Se, and Te), have received a lot of attention for their interesting properties.[1−7] Among them, molybdenum disulfide (MoS_2) is a representative semiconductor and has displayed interesting electrical, mechanical, and optical properties due to a tunable bandgap of 1.2−1.8 eV and a reasonable mobility of 10−200 $cm^2/(V{\cdot}s)$ at room temperature.[8−17] Meanwhile, MoS_2 exhibits good thermal stability and can be grown by chemical vapor deposition in wafer-scale, providing potential compatibility to silicon CMOS processing.[18−21] Although great progress has been made in this field, challenges still remain to achieve high-performance MoS_2 transistors. The mobility of MoS_2 is still far below the theoretically predicted value and is mainly limited by charged impurities and phonon scattering.[22] Studies have shown that the impurities and dielectric environment have an important impact on the electrical and optical properties of two-dimensional materials.[23,24] It has been reported that hexagonal boron nitride (h-BN) is a promising dielectric material for MoS_2, as it can significantly reduce charge traps and greatly enhance the electron field-effect mobility.[25] However, BN flakes are irregularly shaped and are just a few tens of micrometers long, which is a roadblock to their practical use. It has been demonstrated that utilization of high-k dielectric materials as the gate insulator could partially screen charged impurities and improve carrier mobility.[8] Although much work has been devoted to studying the interface between high-k and TMDs, most of it focuses on improvement in the transport properties such as mobility.[26−31] On the other hand, the thermal conductivity (κ) of MoS_2 is very low and excessive heating can lead to poor device performance and decreased lifetimes.[32] Using a high thermal conductivity material as the substrate could facilitate heat dissipation and relieve the self-heating effect.[33] In this work, we fabricated a dual-gated MoS_2 transistor to make a fair comparison of the back interface with SiO_2 and the top interface with Al_2O_3/HfO_2. Carrier transport, pulse *IV*, and low-frequency noise analyses were comprehensively performed to investigate the effect of the dielectric interface on the performance of MoS_2 transistors. The results show that, compared to the SiO_2 substrate, Al_2O_3/HfO_2 could enhance the mobility and improve drain current as well as more effectively eliminate the self-heating effect. Our experimental work demonstrates a promising approach to improve the performance of MoS_2 transistors through substrate engineering.

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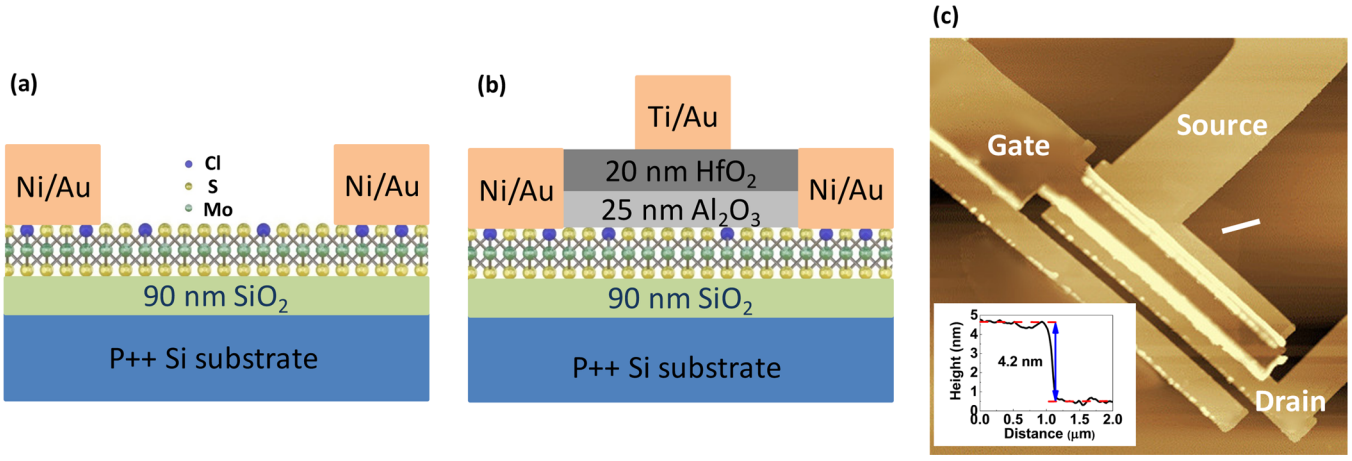


Figure 1. (a) Cross-section schematic view of a back-gate MoS_2 device. (b) Cross-section schematic view of a dual-gate MoS_2 device. (c) AFM images of the MoS_2 device. The MoS_2 flake is 4.2 nm, as measured by AFM.

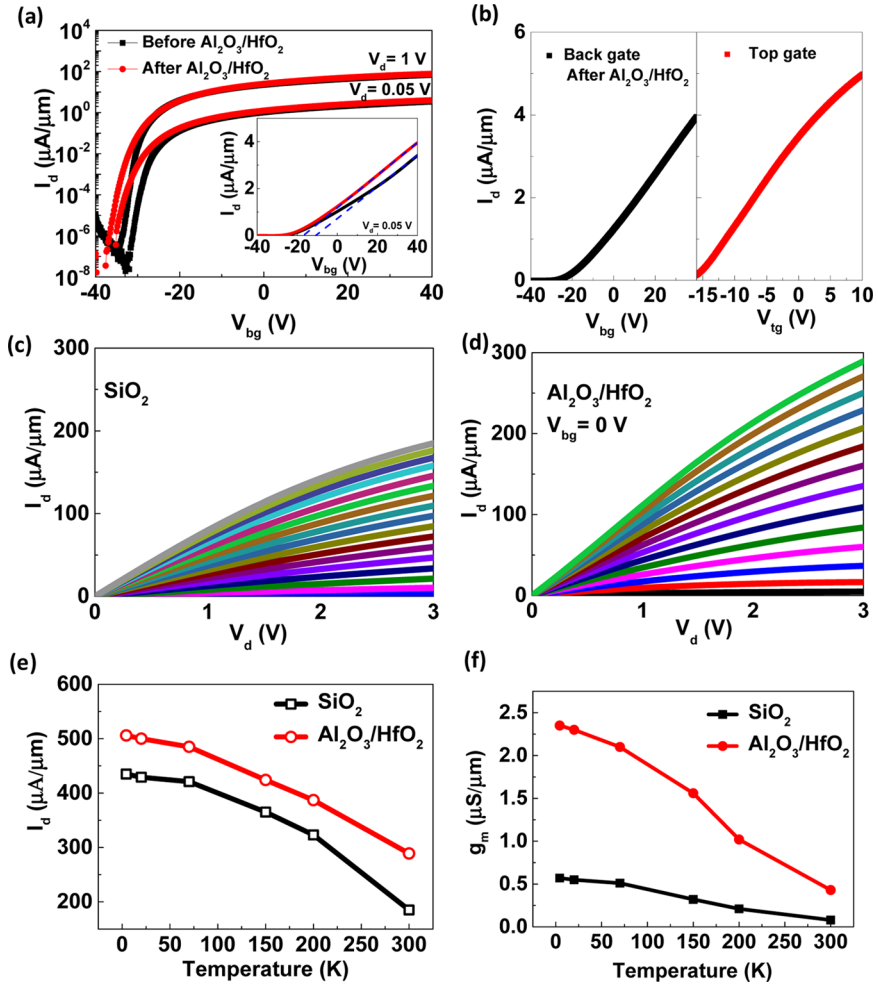


Figure 2. (a) Transfer characteristics of the back-gate MoS_2 device before and after Al_2O_3/HfO_2 capping at 300 K. Inset: the linear coordinated transfer curves. (b) Transfer characteristics of the back-gate and top-gate MoS_2 device at 300 K. (c) Output characteristics of the back-gate MoS_2 device at 300 K. V_{bg} varies from −40 to 40 V in 5 V steps with V_d varying from 0 to 3 V. (d) Output characteristics of the top-gate MoS_2 transistor with V_{bg} fixed at 0 V at 300 K. V_{tg} varies from −16 to 10 V in 2 V steps and V_d varies from 0 to 3 V. (e) Comparison of maximum drain current of the same two devices in (c) and (d) at different temperatures. (f) Comparison of g_m of the same two devices in (c) and (d) at various temperatures.

RESULTS AND DISCUSSION

We start with mechanical exfoliation from commercially available bulk crystals. Figure 1a,b shows the schematic of the MoS_2 device before and after atomic layer deposition (ALD), respectively. We deposited Al_2O_3 at 75 °C to avoid pinholes and nonuniformity in the film. The thickness of MoS_2 used in this work is 4.2 nm, as shown in the inset of Figure 1c, measured by atomic force microscopy (AFM). The channel

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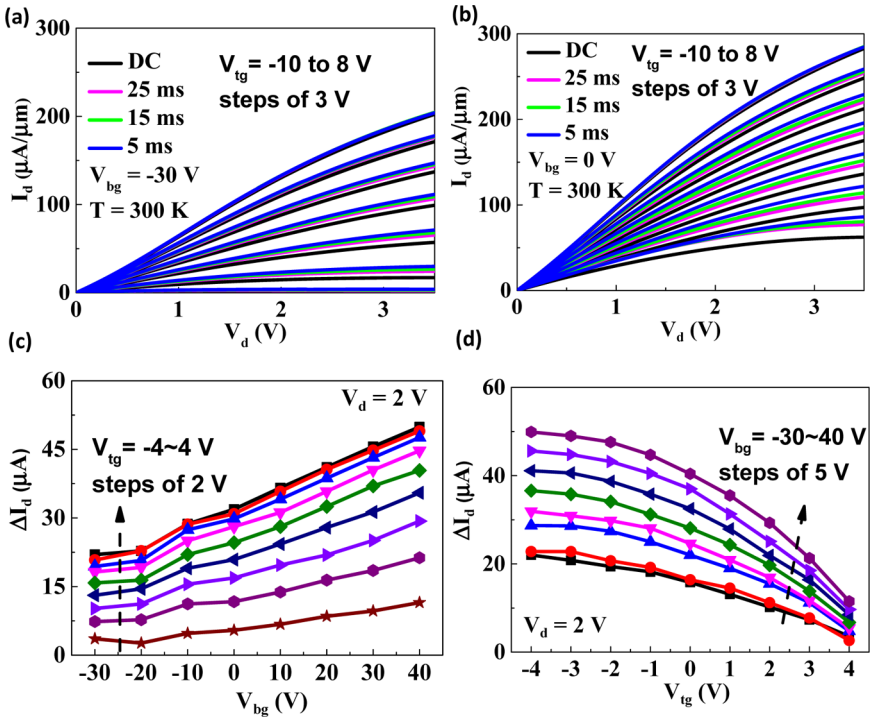


Figure 3. Output characteristics of the top-gate MoS_2 transistor at 300 K with 3 V steps in V_{tg} from −10 to 8 V with comparison between DC measurement and different pulse widths (pulse width = 5, 15, and 25 ms) for (a) V_{bg} = −30 V and (b) V_{bg} = 0 V. (c) ΔI_d as a function of V_{bg} for the same device in (a) at various V_{tg} values with V_d = 2 V. (d) ΔI_d as a function of V_{tg} for the same device in (a) at various V_{bg} values with V_d = 2 V.

length and width of the MoS_2 transistor are 1 and 4.2 μm, respectively. All measurements were carried out under vacuum (<10^{-5} Torr) in a Lakeshore cryogenic probe station using an Agilent parameter analyzer B1500A. The low-frequency $1/f$ noise measurements were carried out by an Agilent B1500A and E5052B with a resistor unit.

We first study the effect of Al_2O_3/HfO_2 capping on the electrical properties of MoS_2 transistors. Figure 2a shows the I_d−V_g curves of back-gate MoS_2 before and after Al_2O_3/HfO_2 deposition. As shown in the inset of Figure 2a, it is clear that the threshold voltage V_{th} shifts toward negative values of around 6 V due to the doping of Al_2O_3/HfO_2,[27,28] suggesting a doping density of 1.4×10^{12} cm^{-2}. Figure 2b shows the transfer characteristics of the back-gate and top-gate MoS_2 transistors at V_d = 0.05 V. The threshold voltage of the back-gate and top-gate FETs are −17.5 and −15.6 V, respectively. The negative V_{th} of the back-gate FET is mainly due to the doping effect from Al_2O_3/HfO_2. A well-behaved output characteristic of a 1 μm-channel-length MoS_2 transistor with a 90 nm SiO_2 gate dielectric is demonstrated in Figure 2c, exhibiting a maximum drain current of 188 μA/μm with V_d = 3 V and V_{bg} = 40 V at 300 K. For the top gate with the Al_2O_3/HfO_2 gate dielectric, the maximum drain current is 290 μA/μm with V_d = 3 V and V_{tg} = 10 V when V_{bg} = 0 V is fixed. The on-current drivability of the top device is higher than that of the back device with the SiO_2 gate dielectric due to the better interface quality as well as a high mobility. Figure 2e shows a comparison of drain current for the back gate and top gate at various temperatures. It is clear that at all temperatures, the drain current of the top gate is larger than that of the back gate, indicating that a high-k dielectric is more effective for improving the performance of MoS_2 transistors. Figure 2f compares peak transconductance (g_m) for the back-gate and top-gate MoS_2 transistors at V_d = 0.05 V from 300 to 20 K. The g_m increases with decreasing temperature for both devices. As expected, at all temperatures, g_m is always higher for the top-gate device. The field-effect mobility μ_{FE} is extracted in the linear region of the transfer characteristics according to the following equations, $\mu_{FE} = g_m/(C_gEW)$ and $(E = (V_d - 2R_cI_d)/L)$, where g_m is the transconductance, C_g is the gate capacitance, E is the transverse electric field in the channel, L and W are the length and width of the channel, respectively, I_d is the drain current, and R_c is the contact resistance.[12] The mobility of the SiO_2 and Al_2O_3/HfO_2 devices at 300 K are 55 and 81 $cm^2/(V{\cdot}s)$, respectively. This improvement in mobility could be attributed to the reduction of Coulomb scattering by dielectric screening,[25] which implies that the Al_2O_3/HfO_2 dielectric has greater potential to realize high-performance transistors with transition minimized differential signaling channel materials.

It is well known that conventional direct current (DC) characterization methods have charge trapping phenomena that occur during long measurements.[34] To minimize the effect of charge trapping and detrapping, pulsed IV measurement is required to obtain the intrinsic electrical characteristics of the MoS_2 transistors.[35,36] DC I_d is compared with I_d measured using different pulse widths, as V_d is swept from 0 to 3.5 V with V_{bg} = −30 V at 300 K as shown in Figure 3a. The pulse width is changed from 5 ms (blue) to 25 ms (magenta). At all V_{tg} values, the drain current increases with shorter pulse width and the DC measurement has the lowest drain current. With increasing V_{bg} to 0 V, the corresponding I_d−V_d curves are shown in Figure 3b. Similar to the case in (a), all pulsed curves are higher than the DC curves. The more traps in the interface, the more drain current degradation ΔI_d.[36] To investigate the interface quality, we define ΔI_d as the difference between the DC and the pulse measurements with 5 ms pulse width at V_d = 2 V. Figure 3c shows ΔI_d versus V_{bg} with various V_{tg} values at 300 K. It is apparent that ΔI_d increases with increasing V_{bg}, and

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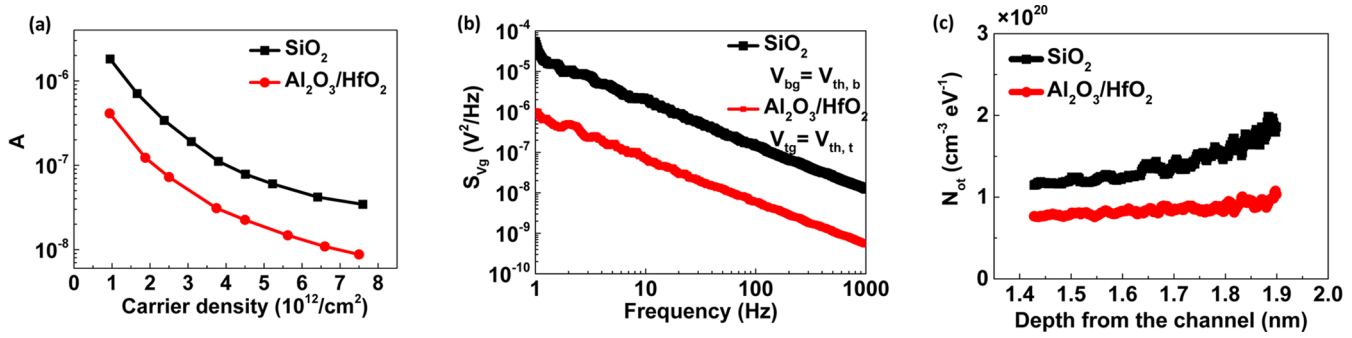


Figure 4. (a) Inverse noise amplitude $1/A$ as a function of carrier density at V_d = 0.2 V. (b) Input gate voltage noise versus frequency for the same two devices in (a). (c) Oxide trap density versus trap depth, derived from the low-frequency noise spectrum at the corresponding threshold voltages $V_{th,t}$ and $V_{th,b}$.

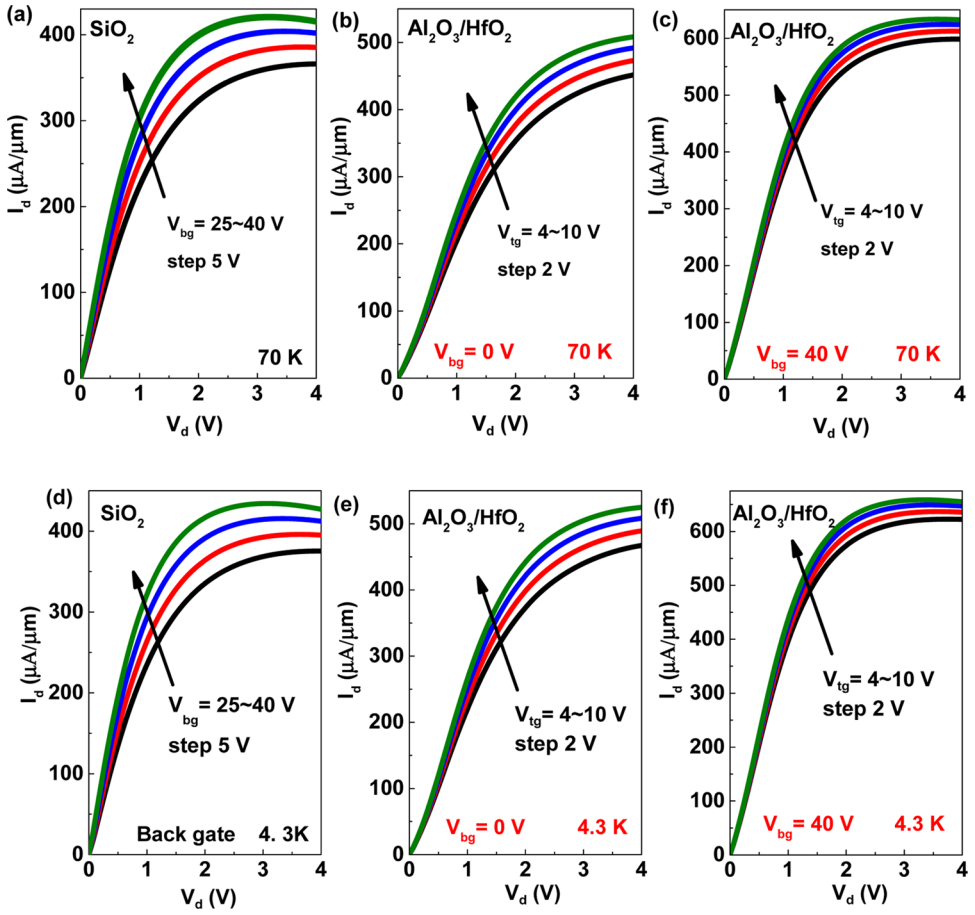


Figure 5. (a) Output characteristics of the back-gate MoS_2 transistor at 70 K. V_{bg} varies from 25 to 40 V in 5 V steps. The I_d–V_d characteristics of the top-gate device at 70 K with (b) V_{bg} = 0 V and (c) V_{bg} = 40 V. V_{tg} varies from 4 to 10 V in 2 V steps. (d–f) The I_d–V_d characteristics of the corresponding devices in (a)–(c) at 4.3 K.

the trend is independent of V_{tg}. Because of the intrinsic n-type nature of MoS_2, electrons in the MoS_2 channel will be close to the Al_2O_3/MoS_2 interface when V_{bg} is negative and will move nearer to the SiO_2/MoS_2 interface for positive V_{bg}. ΔI_d is proportional to the magnitude of traps.[36] Hence, the change in ΔI_d shows that there are fewer traps in the Al_2O_3/MoS_2 interface, indicating a better interface quality compared to that of SiO_2/MoS_2 interface. Figure 3d shows ΔI_d as a function of V_{tg} with different V_{bg} values at 300 K for the same two devices in (c). Obviously, ΔI_d decreases with increasing V_{tg} irrespective of V_{bg}. The explanation is similar to that in (c). For V_{tg} < 0 V, electrons accumulate near to the SiO_2/MoS_2 interface. For V_{tg} > 0 V, electrons approach the Al_2O_3/MoS_2 interface. This result also shows that the interface quality of Al_2O_3/MoS_2 is better than that of SiO_2/MoS_2.

Low-frequency $1/f$ noise is also important because it limits the performance of electronic communication systems and sensors.[37] For example, low-frequency noise determines the noise-to-signal ratio in digital electronic devices and low-frequency noise can be upconverted to phase noise in oscillators.[38] Here, we performed low-frequency noise measurements on the back-gate MoS_2 device and the top-gate MoS_2 device in the linear region with V_d = 0.2 V at 300 K. The noise

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spectra can be quantitatively characterized by $S_{id} = \frac{AI_d^2}{f^\gamma}$, where S_{id} is the current noise power spectral density, f is the frequency, I_d is the current through the device channel, γ is the frequency exponent, and A is the noise amplitude.[39,40] We extract the noise amplitude A using

$$A = (1/N)\sum_{m=1}^{N} f_m S_{Im}/I_m^2 \quad (1)$$

where S_{Im} and I_m are the noise spectral density and drain current measured at m different frequencies.[41] Figure 4a shows the normalized A values of the back-gate and top-gate MoS_2 transistors as a function of carrier density at 300 K. The noise amplitude from the top gate improves by more than 40% over that from the back gate for the same carrier density, indicating that the interface quality of Al_2O_3/MoS_2 is better than that of SiO_2/MoS_2. To evaluate quantitatively the interface property, we estimated the oxide trap density (N_{ot}) using input-referred voltage noise spectral density S_{vg} at the threshold voltage (V_{th}), as shown in Figure 4b. It can be seen that the top-gate exhibits reduced noise, implying a lower oxide trap density. Previous studies have demonstrated that the dominant noise mechanism of few-layer MoS_2 transistors is number carrier fluctuation.[12,42,43] It is therefore possible to evaluate the oxide trap density using the following formula.[44,45]

$$S_{vg} = \frac{N_{ot}kTq^2}{\alpha f W L C_g^2} \quad (2)$$

where q is the element charge, kT is the thermal energy, and α is the tunneling constant ($\alpha = 10^8$ cm^{-1} is used here). The trap depth z at which the traps in the oxide are measured is derived from the noise frequency using[46]

$$z = \frac{1}{\alpha}\ln\frac{1}{2\pi f\tau_0} \quad (3)$$

where the time constant τ_0 is often taken as 10^{-10} s. On the basis of eqs 2 and 3, we extracted the oxide trap density profile for the back-gate and top-gate transistors. The depth z of both devices is 1.43−1.90 nm with the noise measurements in the frequency range from 10 to 1000 Hz. Figure 4c shows the oxide trap density as a function of depth for both devices. The N_{ot} of 7.7×10^{19}−1.1×10^{20} cm^{-3} eV^{-1} for the top-gate device is lower than that of the back-gate device (i.e., $(1.2–2.0) \times 10^{20}$ cm^{-3} eV^{-1}). This result is in agreement with our observation from pulse IV results and is consistent with previous reports.[47,48]

To further demonstrate the advantages of Al_2O_3/HfO_2, we compared the output characteristics of the back-gate and top-gate MoS_2 transistors at 70 and 4.3 K. Figure 5a depicts the well-behaved output characteristic of the back-gate device using SiO_2 substrate with V_d sweeping from 0 to 4 V for four different V_{bg} values at 70 K. At V_{bg} = 25 V, the maximum drain current is 375 μA/μm, and the curve exhibits current saturation. With increasing V_{bg} further, the drain current increases accordingly and even shows signs of the negative differential resistance (NDR) effect when I_d is larger than 400 μA/μm due to the self-heating effect, which is consistent with previous studies.[12,49] The saturation current degradation $\Delta I/I_{sat}$ with V_{bg} = 40 V at 70 K is 1.2%. This is predictable because both MoS_2 and SiO_2 have a low thermal conductivity, indicating poor heat transport behavior.[32,33] The local heating of MoS_2−SiO_2 interfaces could be even more pronounced compared to that of Si−SiO_2 interfaces in modern silicon-on-insulator structures, which could be an important issue for high-performance MoS_2 transistors with scaling.[50] For efficient heat dissipation from the MoS_2 channel, substrates with high thermal conductivity (κ) are desirable, e.g., Al_2O_3 (κ = 35 W/(m·K)), instead of the commonly used SiO_2 (κ = 1.4 W/(m·K)).[33] Figure 5b shows the output characteristics of the top-gate MoS_2 transistors at 70 K with top-gate biases and V_{tg} ranging from 4 to 10 V in 2 V steps and V_{bg} fixed at 0 V. We did not observe any NDR effect, even with 18% improvement in drain current over that of the device in (a), indicating better interface passivation and efficient heat dissipation using Al_2O_3 with high thermal conductivity. With increasing V_{bg} to 40 V, the drain current is over 600 μA/μm without any NDR effect, as shown in Figure 5c. When decreasing the temperature down to 4.3 K, the drain current increases accordingly, as shown in Figure 5d−f. The saturation current degradation $\Delta I/I_{sat}$ of the back-gate device with V_{bg} = 40 V at 4.3 K is 1.6%, as shown in Figure 5d, which is larger than that of 1.2% at 70 K in (a). This confirms that the NDR effect is more pronounced at higher drain current due to the self-heating effect.[51] At V_{bg} = 40 V and 4.3 K, the drain current of the top-gate device is larger than 620 μA/μm, as shown in Figure 5f, up to 660 μA/μm at V_{tg} = 10 V. When the drain current is greater than 635 μA/μm, we observed signs of the NDR effect again. $\Delta I/I_{sat}$ with V_{tg} = 10 V and V_{bg} = 40 V at 4.3 K is 0.5%, which is three times smaller than that in (d). This result shows that Al_2O_3/HfO_2 is more efficient for removing self-heating. It should be noted that the NDR effect is strongly related to the width and channel length of MoS_2 transistors, which determines the area of dissipation during device operation. With further scaling, the NDR phenomena are more easily observed due to the increased heat generation and reduced area for heat dissipation.

CONCLUSIONS

In conclusion, we report a high-performance MoS_2 transistor. We systematically compare the effect of Al_2O_3/HfO_2 and SiO_2 substrates on the mobility, drain current, oxide trap density, and NDR of MoS_2 devices. Benefiting from the high dielectric constant and high thermal conductivity of Al_2O_3/HfO_2, the MoS_2 transistor with Al_2O_3/HfO_2 exhibits a high mobility of 81 cm^2/(V·s) and a lower oxide trap density. The maximum drain current of the top-gate device with a 1 μm-channel-length is up to 660 μA/μm at 4.3 K. Meanwhile, the Al_2O_3/HfO_2 substrate reduces the self-heating effect. Our work provides a viable path to improve the device performance of TMD-based FETs.

METHODS

By standard scotch-tape techniques, few-layer MoS_2 was transferred to heavily doped silicon substrate, which was capped with 90 nm SiO_2. Then, it was soaked in undiluted 1,2-dichloroethane at room temperature for more than 12 h. Source/drain regions were defined by e-beam lithography, followed by a Ni/Au metallization of 15/45 nm with an e-beam evaporator. The gap between source and drain was 1 μm. After the device was first measured with a back gate only, the top-gate dielectric was grown by ALD. First, 25 nm of Al_2O_3 was deposited with trimethylaluminum and H_2O as precursors at 75 °C and then 20 nm of HfO_2 was in situ deposited using tetradimethylaminohafnium and H_2O as precursors at 150 °C. The dielectric constant of Al_2O_3 and HfO_2 are 6 and 12, respectively. The top gate was defined with e-beam lithography and Ti/Au of 15/45 nm was used as the top gate. No annealing was performed during the process.

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Notes

The authors declare no competing financial interest.

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